

Supplementary Material: Validated Read-Only Geometric Stress Testing of Native Assignment States in Aerial Tiny-Object Detection

Yede Liu^{a,1}, Fang Wang^a, Jun Li^{a,*}

^aCollege of Physics and Electronic Engineering, Chenglong Campus, Sichuan Normal University, No. 1819, Section 2, Chenglong Avenue, Longquanyi District, Chengdu, Sichuan 610101, China

1. Purpose, evidence hierarchy, and interpretation rules

This Supplementary Material expands the frozen evidence used in the main manuscript. It does not introduce new training or post hoc outcome selection. The evidence hierarchy is: (i) `evidence_freeze/final_evidence_v2`; (ii) the final evidence ledger; (iii) historical factual records; and (iv) manuscript prose. The final v2 manifest SHA-256 is

4f626da49dcd7ee87ffd33311e9c08c73629e2b7b4cb0b2c7e74501a7a5a1ca9.

The associated validation passed all 16 checks. Twenty-four evidence entries were admitted, five records were explicitly excluded, and no evidence item remained pending.

The principal endpoint estimates are inverse-inclusion weighted. Unless stated otherwise, 95% intervals use 5,000 bootstrap replicates that resample image clusters within the frozen outcome-blind sampling strata. Fixed one-pixel and equivalent-side replay are distinct measurement contracts and are never pooled. First-divergence labels localise the earliest instrumented native state difference; they are not causal effects. The prospective lockbox produced no normative verdict. Corrected post-access results are exploratory only.

Table S1: Evidence-status vocabulary used throughout the paper.

Status	Meaning
Qualified instrument	Analytical oracles, mutation tests, invariants, native parity, and independent-reference checks passed.
Discovery estimate	Frozen outcome-blind discovery-sample estimate with prespecified weighting and cluster bootstrap.
Sensitivity/replication	Read-only analysis under another contract, dataset, checkpoint, or state definition.
Negative boundary	Prespecified analysis did not establish the claimed utility or treatment effect; not an equivalence claim.
No normative verdict	The single lockbox analyzer invocation ended before any preregistered estimate or decision was written.
Post-access exploratory	Corrective estimate computed after outcome access; formal status is permanently null.

2. Detector, dataset, sampling, and replay contracts

The principal contract was YOLO26s trained on AI-TOD-v2 with native O2O and O2M assignment branches. VisDrone supplied cross-dataset sensitivity, and a compatible YOLOv10-S implementation supplied detector-contract replication. All scientific replays used frozen checkpoints and frozen decoded outputs. Replay changed only the focal GT centre; it did not change image pixels, object width or height, candidate grids, parameters, features, other GTs, or detector inference.

The AI-TOD-v2 discovery bundle contained 300 outcome-blind stratified validation images. Sampling weights were inverse image-inclusion probabilities. The principal equivalent-side displacement was $\delta_g =$

*Corresponding author: Jun Li. Email: lijun2022@sicnu.edu.cn.

Email addresses: liuyede0718@gmail.com (Yede Liu), fang66320@gmail.com (Fang Wang), lijun2022@sicnu.edu.cn (Jun Li)

¹ORCID: Yede Liu, [0009-0004-1760-860X](https://orcid.org/0009-0004-1760-860X).

$0.0625\sqrt{w_g h_g}$, with $\kappa \in \{0.03125, 0.0625, 0.125\}$ used for stress-magnitude sensitivity. Invalid boundary shifts were skipped rather than clipped.

The downstream coverage-based-miss outcome indicates that no frozen same-class prediction among the prespecified outputs overlaps the focal GT by at least 0.5. Predictions are not consumed one-to-one; the outcome is therefore not the standard evaluator false negative.

3. Instrument qualification and numerical parity

The qualification suite contained six analytical oracle cases, seven property invariants, and six prespecified mutations. All oracles and invariants passed, and every mutation was detected. Native execution, production tracing, and the independent NumPy reference produced zero mismatch across 300 base states and 5,251 deterministic directional states.

Continuous-boundary qualification used 30 frozen images and 4,408 trajectories. The independent NumPy implementation reproduced all production event types and radii with zero difference and verified one network forward per image. Float32 native execution was normative; exact ties, eligibility comparisons, clipped-CIoU zero states, ordering, margin epsilon, and boundary brackets followed the frozen numerical contract.

Table S2: Instrument qualification summary.

Check	Cases/states	Mismatches	Result
Analytical oracles	6	0	pass
Property invariants	7	0	pass
Prespecified mutations	6	0 undetected	pass
Native/tracer/NumPy parity	5,551 states	0	pass
Continuous-boundary reference	4,408 trajectories	0	pass
Forward-call contract	30 images	0 excess calls	pass

4. Endpoint stress-contract results

4.1. Primary fixed and equivalent-side contrasts

Table S3: AI-TOD-v2 primary scale contrasts in percentage points. Positive values denote higher 8–16 px than 16–32 px fragility.

Stress contract	O2O scale gap	O2M scale gap	Paired gap
Fixed 1 px	14.37 [10.84, 17.78]	1.42 [-3.18, 6.31]	12.95 [6.52, 19.16]
Equivalent-side $\kappa = 0.0625$	0.13 [-3.41, 3.15]	-17.45 [-22.72, -11.71]	17.58 [11.06, 23.98]

4.2. Magnitude and dataset sensitivity

Table S4: Equivalent-side stress sensitivity. Values are 8–16 minus 16–32 px contrasts in percentage points with 95% stratified image-cluster bootstrap intervals.

Dataset	κ	O2O	O2M	O2O–O2M
AI-TOD-v2	0.03125	1.34 [-0.72, 3.10]	-16.85 [-22.30, -11.00]	18.19 [12.48, 23.83]
AI-TOD-v2	0.06250	0.13 [-3.41, 3.15]	-17.45 [-22.72, -11.71]	17.58 [11.06, 23.98]
AI-TOD-v2	0.12500	8.91 [4.21, 13.55]	-8.36 [-11.33, -5.41]	17.27 [11.43, 23.45]
VisDrone	0.03125	5.14 [3.93, 6.36]	-29.65 [-31.73, -27.59]	34.79 [32.19, 37.42]
VisDrone	0.06250	11.04 [9.33, 12.70]	-26.21 [-27.85, -24.57]	37.25 [34.73, 39.85]
VisDrone	0.12500	19.97 [17.73, 22.25]	-16.01 [-17.17, -14.84]	35.98 [33.39, 38.56]

Absolute O2O response varied with stress magnitude and dataset; the paired branch direction was positive at all six tested combinations. This supports a contract-dependent branch difference rather than a universal O2O scale law.

Table S5: Paired O2O-minus-O2M scale contrasts across the complete physically archived trajectory. Epoch 0 denotes the checkpoint written after the first completed training epoch. No intermediate archive existed for epochs 100–280.

Epoch	mAP _{50:95}	Fixed pp	Equivalent-side pp	Common-valid GTs (fixed/norm.)
0	0.112	13.42 [7.57, 20.12]	18.41 [11.97, 25.51]	6343/6342
20	0.191	13.06 [6.68, 19.71]	19.27 [12.92, 26.00]	6394/6393
40	0.226	15.05 [8.71, 21.35]	21.35 [14.38, 28.38]	6392/6391
60	0.243	15.09 [10.23, 20.05]	21.00 [15.65, 26.48]	6387/6386
80	0.254	14.16 [8.21, 20.12]	19.31 [13.93, 25.03]	6394/6393
300	0.280	11.84 [5.70, 18.10]	17.34 [11.22, 23.38]	6374/6373

4.3. Archived-stage and detector-contract sensitivity

YOLOv10-S equivalent-side replay yielded O2O +0.94 pp [−1.85, 3.61], O2M −23.50 pp [−28.18, −18.90], and paired +24.44 pp [18.89, 30.06]. This is replication under two compatible detector contracts, not architecture invariance.

5. Eligibility, first-divergence, and pre-resolution controls

5.1. Eligibility-lock and eligibility-stable estimands

Holding base eligibility fixed reduced the fixed O2O scale gap from 17.10 to 3.52 pp, a reduction of 13.58 pp [11.21, 15.87] or 79.4% [70.5%, 91.9%]. The normalized gap moved from 2.95 to −0.20 pp, a reduction of 3.15 pp [1.42, 4.83]. The structural counterfactual therefore removed most of the fixed-pixel scale response.

The eligibility-stable observational subset retained 6,217/7,434 fixed objects and 6,999/7,432 equivalent-side objects. Object-level fixed fragility was 7.95% [6.57%, 9.43%] for 8–16 px and 4.59% [2.65%, 7.01%] for 16–32 px, a 3.36 pp [1.11, 5.48] contrast. The equivalent-side contrast was 1.05 pp [−1.19, 2.96]. Direction-level contrasts were 1.02 pp [0.19, 1.74] and 0.02 pp [−0.86, 0.76], respectively. Because stability is conditioned after replay, these are selected estimands.

5.2. Corrected first-divergence anatomy

Table S6: Mutually exclusive first-divergence shares among directional O2O identity exchanges. Geometry clipping is an overlapping diagnostic and is not included in the 100% sum.

Pathway	Fixed 1 px	Equivalent side	Equivalent minus fixed
Eligibility boundary	67.09 [63.02, 71.07]	51.01 [46.06, 55.94]	−16.09 [−18.92, −13.34]
Top- <i>k</i> transition	0.27 [0.05, 0.54]	0.48 [0.17, 0.85]	0.21 [−0.04, 0.48]
Conflict reassignment	2.51 [1.07, 4.12]	3.14 [1.61, 4.93]	0.64 [−0.71, 1.96]
Within-set geometry	30.13 [26.14, 34.36]	45.37 [40.62, 50.26]	15.24 [12.69, 17.83]
Active disappearance	0	0	0
Compound/other	0	0	0
Directional flip rows	2,194	1,241	n/a
Contributing images	201	201	201 paired
Geometry-clipping event	44.72 [41.18, 48.73]	31.8 (overlapping)	n/a

Frozen outputs make the stress-induced classification term zero by construction. The useful decomposition is therefore between continuous clipped-CIoU movement among comparable candidates and discrete eligibility, Top-*k*, and conflict-state transitions. Algebraic residual was below 2×10^{-7} and is treated as an implementation check, not a scientific discovery.

5.3. Pre-resolution O2O control

Under fixed replay, the pre-conflict Top-7 winner scale gap was 17.18 pp [13.47, 20.80], versus 17.10 pp [13.37, 20.75] for the final loss-active identity. Their difference was −0.077 pp [−0.367, 0.313]. Under equivalent-side replay, the corresponding gaps were 3.34 pp [−0.23, 6.52] and 2.95 pp [−0.76, 6.20], a −0.395 pp [−0.986, 0.034] difference. Object-level fragility-definition disagreement was 0.578% under both contracts. Every observed directional disagreement coincided with an instrumented conflict-stage difference; this localises but does not identify a causal resolver contribution.

6. Continuous assignment-boundary geometry

The formal analysis contained 29,724 directional trajectories from 7,431 focal GTs. Equivalent-side radius was scanned forward through $r_{\max} = 0.25$ with a 1/64 coarse step and a 1/512 fine step. The search did not assume monotonicity and retained return transitions. A first boundary beyond the scan range was right-censored; an earlier pathway change could be a competing censor for the fixed-pair rank-crossing estimand.

Table S7: Restricted mean stability radius (RMSR) through $\tau = 0.125$.

Quantity	Point	95% CI
O2O RMSR	0.11992	[0.11940, 0.12038]
O2M assigned-positive RMSR	0.10055	[0.09895, 0.10210]
Tiny O2O–O2M RMSR	0.01724	[0.01565, 0.01876]
Small O2O–O2M RMSR	0.03282	[0.02904, 0.03614]

For O2O, the observed boundary incidence by τ was 9.03% [8.26%, 9.90%] for 8–16 px and 6.39% [4.61%, 8.49%] for 16–32 px. For O2M assigned-positive Top-1, it was 32.66% [31.19%, 34.14%] and 48.15% [44.79%, 51.31%], respectively. The RMSR contrast O2O minus assigned-positive O2M was 0.01724 [0.01565, 0.01876] for 8–16 px and 0.03282 [0.02904, 0.03614] for 16–32 px.

For fixed-pair rank-crossing radius ρ_R , 1,592 trajectories were observed crossings, 7,437 were competing-censored, 16,551 were right-censored, and 4,144 were undefined. Undefined reasons were 676 base-active missing, 56 base order not active over runner, and 3,412 base runner missing.

7. O2M Rank–Set and same-stride results

Table S8: Tiny-minus-small O2M Rank–Set contrasts in percentage points.

Contract	Δ Jaccard	Δ retention	Δ Top-1 flip	Δ rank-only share
Fixed 1 px	-4.19	-2.26	-2.38	-10.10
Equivalent-side	5.91	4.93	-12.80	22.50

Under equivalent-side replay on AI-TOD-v2, the weighted mean native positive count was 3.07 [2.96, 3.17] for 8–16 px and 6.54 [6.29, 6.81] for 16–32 px, a difference of +3.47 [3.20, 3.78]. Base rank margin was 0.220 [0.196, 0.245] and 0.116 [0.102, 0.131], a 16–32 minus 8–16 difference of -0.104 [-0.131 , -0.079]. Same-stride local transitions accounted for 55.0% [45.8%, 63.9%] of the reverse directional gap; same-stride nonlocal and cross-stride transitions accounted for 21.6% and 23.3%.

The strict post-replay same-stride subset retained 3,835 paired GTs. Its O2O contrast was +10.23 pp [7.30, 13.32], O2M was -0.12 pp [-6.83 , 5.78], and the paired difference was +10.35 pp [3.54, 18.17]. This sensitivity rules out an explanation based solely on cross-stride switching but remains vulnerable to selection induced by post-replay conditioning.

8. Margin construct, shape, and held-out diagnostic value

Among the 1,592 observed fixed-pair crossings, native margin correlated with ρ_R at $\rho = 0.628$ [0.584, 0.670]. Its correlation with observed eligibility-boundary radius was 0.153 [0.114, 0.194]. The difference was 0.475 [0.405, 0.540]. These correlations condition on observing the relevant boundary and are not censoring-adjusted population associations.

Adding margin after log area, log O2M positive count, log native active score, and active CIoU changed held-out coverage-based-miss ROC-AUC by -0.00100 [-0.00484 , 0.00036], PR-AUC by +0.00366 [-0.00405 , 0.01197], log loss by +0.00041 [-0.00362 , 0.00534], and Brier score by +0.000012 [-0.000375 , 0.000496]. Thus, rank-separation construct validity did not translate into stable incremental prediction under the tested model sequence.

9. Locality, audit budget, and secondary negative boundaries

9.1. Non-focal spillover

The 300-image spillover analysis contained 3,549,250 focal–non-focal pair-events. Fifty-two non-focal identity changes were observed. The overall weighted pair rate was 0.001465% [0.000824%, 0.002383%]. All

Table S9: Held-out effect of replacing the linear margin term with a restricted cubic spline. Values are spline minus linear model.

Outcome	Metric	Difference	95% CI
Fragility	ROC-AUC	-0.00756	[-0.01411, -0.00138]
Fragility	PR-AUC	-0.00415	[-0.00813, -0.00109]
Fragility	Log loss	-0.00090	[-0.00260, 0.00075]
Coverage-based miss	ROC-AUC	-0.00057	[-0.00122, 0.00002]
Coverage-based miss	PR-AUC	-0.00541	[-0.01298, -0.00027]
Coverage-based miss	Log loss	-0.00060	[-0.00794, 0.00526]

events occurred in 5,679 pairs sharing a native pre-conflict candidate claim, for a weighted rate of 0.9159% [0.5787%, 1.3248%]. No event was observed among 3,543,571 nonsharing pairs. Zero observations do not establish absence, independence, or equivalence.

9.2. Reduced audit budget

Table S10: Outcome-blind subsampling error for the normalized paired contrast; 1,000 repeated stratified subsamples per budget.

Budget	Median error	P95 error	Sign recovery	Order recovery
25	6.39	18.38	97.5%	97.5%
50	4.52	13.63	99.6%	99.6%
100	3.43	8.41	100.0%	100.0%
150	2.45	6.13	100.0%	100.0%
200	1.74	4.60	100.0%	100.0%

At 100 images, all 1,000 repeats recovered the sign and branch ordering of the full normalized paired contrast, while median and 95th-percentile absolute error remained 3.43 and 8.41 pp. The result supports dominant-signature screening, not precise effect estimation or a universal sample-size threshold.

9.3. Previously tested interventions

The commitment, relaxation, matched-selector controls, and detached O2M-evidence Resolver are retained as secondary negative boundaries. They were not promoted to principal contributions because the paper is a measurement study and the tested operations do not span the space of possible assignment interventions.

Table S11: Frozen intervention outcomes. Values are $AP_{50:95}$ unless stated otherwise.

Operation	AP	Difference from Native	Interpretation
Native checkpoint	0.24957	n/a	reference
Margin commitment	0.24846	-0.111 pp	no efficacy established
Random matched selector	0.24841	-0.116 pp	matched control
Margin relaxation	0.24739	-0.218 pp	no efficacy established
Score matched selector	0.24743	-0.214 pp	matched control
Detached O2M Resolver	0.22803	-2.154 pp	negative boundary

Controlled selector evidence and the O2M transfer boundary

matched controls · size deltas · transfer gate

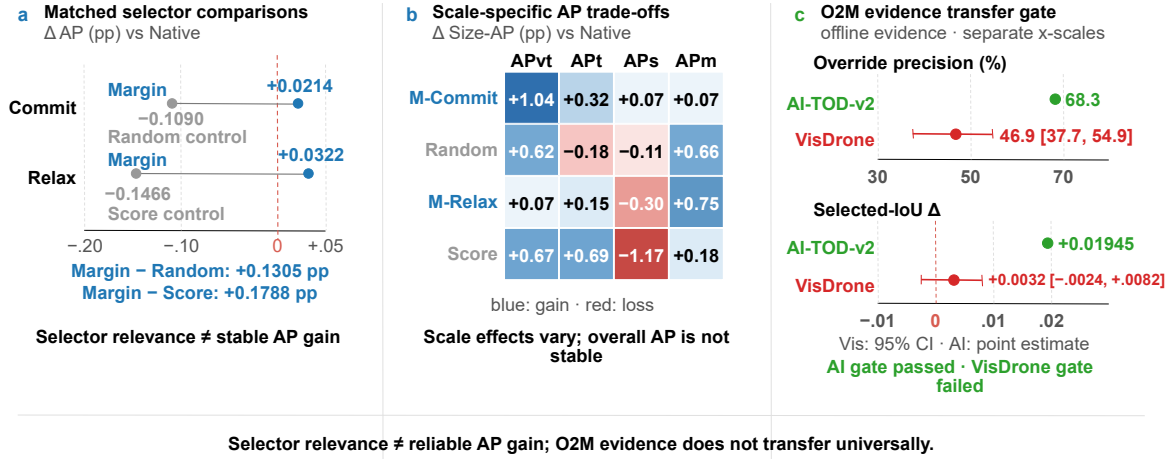


Figure S1: Secondary controlled-intervention evidence. The matched controls separate selector relevance from treatment efficacy. The figure is retained for transparency and does not imply that all assignment interventions are ineffective.

AI-TOD-v2 passed the offline evidence-transfer gate: coordinate correspondence was 100% on 1,500 sampled objects, low-margin override precision was 68.3% [59.6%, 77.1%], and evidence-selected IoU exceeded Native by 0.01945 [0.01012, 0.03084]. VisDrone also had 100% correspondence, but precision was 46.9% [37.7%, 54.9%] and its IoU difference was 0.0032 [-0.00236, 0.00816]; the overall gate failed, so no VisDrone Resolver was trained.

10. Qualitative provenance

Read-only identity replay under a 1 px focal-GT shift

AI-TOD-v2 · three focal GTs · frozen checkpoint · no postprocessing

Source image



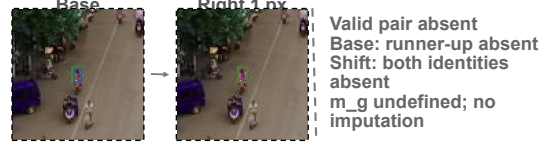
a Stable identity



b Fragile identity



c Undefined competition



green: focal GT · blue: active · orange: runner-up

Figure S2: Frozen qualitative scene-to-crop correspondence. Colored focal-GT boxes are audit annotations, not detector outputs. Crop identities, active candidates, runners, and labels were deterministically derived from the frozen replay records.

The stable, fragile, and undefined examples were selected from the frozen qualitative manifest before final layout. They illustrate the native state definitions and do not estimate prevalence. Undefined examples remain excluded from continuous margin analysis without imputation.

11. Prospective lockbox chronology and exploratory correction

The lockbox design selected a disjoint outcome-blind 300-image AI-TOD-v2 sample and froze the implementation, numerical contract, sampling, taxonomy, bootstrap, checkpoint hashes, and hierarchical Tier-A/Tier-B decision rules before outcome access. Four sealed extraction roles completed. The single permitted analyzer invocation then wrote the irreversible outcome-access receipt and terminated before any H1–H5 estimate or decision because support assertions were evaluated before the specified zero-cluster and common-support operations. It was not rerun.

Accordingly, no hypothesis passed, failed statistically, or was found nonsignificant: no normative verdict exists. The corrective analysis below was specified after outcome access, retains formal status `null`, and is reported only to disclose direction and support handling.

Table S12: Post-access corrective contrasts. These are exploratory estimates, not preregistered verdicts.

Analogue	Point	Exploratory 95% CI	Formal status
H1	16.69	[12.47, 20.74]	invalidated; not retested
H2	-16.99	[-20.62, -13.34]	invalidated; not retested
H3	14.00	[12.51, 15.60]	invalidated; not retested
H4	21.15	[18.54, 23.44]	invalidated; not retested
H5	19.31	[16.53, 21.94]	invalidated; not retested

The correction retained all 300 image clusters and used the 7,243-key fixed/normalized common-valid intersection; three fixed-only keys were disclosed. Directional agreement with the preregistration cannot recover prospective status. The episode is treated as a support-contract validation failure and motivates pre-outcome testing of zero-row clusters, merge order, and common-support assertions.

12. Reproducibility inventory

The public repository is https://github.com/liuyede447/020_identity_audit. The frozen internal package retains run arguments, selected-image IDs, image inclusion probabilities, per-object and per-direction derived tables, 5,000-replicate bootstrap summaries, figure source data, checkpoint SHA-256 identifiers, lockbox receipts, formal failure records, and independent-validation reports. Raw benchmark images, third-party annotations, model binaries, and raw prediction dumps are not redistributed.

Table S13: Minimum file-level audit trail for the final manuscript.

Component	Frozen record
Evidence authority	<code>evidence_freeze/final_evidence_v2</code> and its manifest/SHA-256 files
Numerical manuscript values	<code>manuscript/generated_final_v2</code>
Figure sources	<code>manuscript/figures/final_v1</code> ; each result figure has PDF, SVG, 600 dpi PNG, script, and validation JSON
All Goal-mode results	<code>reports/GOAL_MODE_ALL_RESULTS_20260901.md</code> with validation JSON
Prospective record	Immutable access receipt, failed analyzer log, and no-verdict record
Public materials	Audit code, manifests, derived tables, bootstrap summaries, and figure source data in the project repository